

WASSIM BOUSBIA

Cairo, Egypt +201013638673
wassim@aucegypt.edu ◇ bousbiaw@g.cofc.edu
linkedin: Wassim Bousbia ◇ github.com/wassimbousbia

EDUCATION

The American University in Cairo (AUC) <i>Bachelor of Science in Data Science</i>	Sep 2023 – Present <i>Cairo, Egypt</i>
• Awarded Tomorrow's Leaders Undergraduate Scholarship. • Relevant coursework: ML, Statistical Modeling, Data Structures & Algorithms, Linear Algebra, Probability & Statistics	
College of Charleston (CofC) <i>B.S. Data Science — Exchange Semester</i>	Jan 2026 – May 2026 <i>South Carolina, USA</i>
Mohand Mokhbi High School <i>Baccalaureate Certificate, Mathematics Branch</i>	2019 – 2022 <i>Algiers, Algeria</i>
• GPA: 17.17/20 — Mention: Good	

RESEARCH & PROJECT EXPERIENCE

South Carolina Real Estate Forecasting <i>Statistical Modeling & Simulation, College of Charleston</i>	Feb 2026
• Engineered an automated ML pipeline in Python processing a 2.6GB+ dataset (13 years); handled ingestion, cleaning, feature engineering, and simulation outputs • Implemented Monte Carlo simulations (10,000 paths, 36-month horizon) to model probabilistic price distributions • Quantified predictive uncertainty via VaR and downside risk metrics; compared regression and time-series model configurations	
Credit Card Fraud Detection <i>Anomaly Detection & Imbalanced Classification, The American University in Cairo</i>	Aug 2025
• Built a reproducible, modular ML pipeline with automated preprocessing, threshold tuning, and documented evaluation methodology • Applied SMOTE resampling to address severe class imbalance; optimized decision thresholds for precision-recall trade-off • Trained and benchmarked Logistic Regression, Random Forest, and Gradient Boosting; model selection via ROC-AUC and F1; analyzed feature importance to identify key discriminative signals	
Multi-Stock Forecasting System <i>Time-Series Regression Research, The American University in Cairo</i>	May 2025
• Designed modular, reusable regression pipelines for multi-asset time-series prediction; enforced strict out-of-sample validation to prevent data leakage • Engineered rolling statistical features (moving averages, volatility windows) to capture temporal structure; built structured hyperparameter tuning and experiment comparison framework	
Gold Price Prediction — XGBoost Time-Series Model <i>Belton AI Hackathon</i>	Jun 2024
• Trained an XGBoost regression model achieving 89.08% accuracy for next-day forecasting; engineered lag features and applied cross-validation for robust generalization • Packaged experiment results into structured comparison reports; documented full methodology for reproducibility	

INDUSTRY EXPERIENCE

Data Analyst - Freelance <i>Remote</i>	Jan 2025 – Present
• Delivered +20 DA and ML projects for multiple clients, including governmental authorities; built pipelines covering data cleaning, EDA, modeling, clustering and visualization reporting.	
Internal Public Relations & Stakeholder Management <i>AUC International Symposium, The American University in Cairo</i>	Jan – Jun 2025
• Managed stakeholder relationships across 8+ university departments and external partners; prepared structured budget reports for senior leadership • Coordinated logistics for a 300+ participant international event; managed a \$20,000 cross-functional budget	
Action Committee Member <i>Tomorrow's Leaders Annual Symposium</i>	Jan – Feb 2024
• Collected and analyzed operational data from 42 NGOs to evaluate program impact and inform professional development recommendations	

AWARDS & CERTIFICATES

• Tomorrow's Leaders Undergraduate Scholarship , The American University in Cairo	2023–Present
• Research Nomination: students honored for outstanding papers	2024
• Finalist & Certificate of Participation: Belton AI Hackathon ◇ Deloitte Datathon ◇ Finomics Competition ◇ Algerian Youth Leadership Program Alumni ◇ Algerian MUN Delegate	

SKILLS

Programming: Python (primary), R, SQL, Java, C++
ML / AI: scikit-learn, XGBoost, Pandas, NumPy, TensorFlow, PyTorch, Clustering, Anomaly Detection
Generative AI & NLP: Transformers (Hugging Face), LLMs, RAG, Prompt Engineering
Data & Visualization: Feature Engineering, EDA, Statistical Modeling, Matplotlib, Plotly, Seaborn
Tools & Cloud: Git, GitHub, Docker, Jupyter Notebook, VS Code
Other: Excel, Power BI
Languages: English (Fluent), French (Fluent), Arabic (Native)